$$A = \left\{ x \in \mathbb{R} : \frac{2x-1}{x-3} \geq 1 \right\}$$

$$\frac{2x-1}{x-3} \geq 1$$

$$\frac{2x-1}{x-3} - \frac{1}{1} \geq 0$$

$$\frac{2x-1-(x-3)}{x-3} \geq 0$$

$$\frac{2x-1-x+3}{x-3} \geq 0$$

$$\frac{x+2}{x-3} \geq 0$$

OBS.:

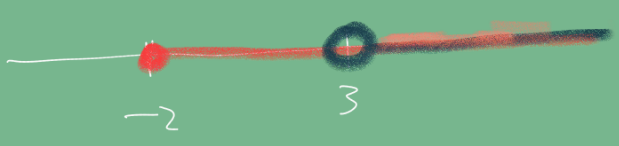
$\frac{A}{B} \geq 0$ → $A \geq 0$ y $B > 0$

$\frac{A}{B} \geq 0$ → $A \leq 0$ y $B < 0$

PLANTEAMOS CASOS:

1) $x+2 \geq 0$ y $x-3 > 0$

$x \geq -2$ y $x > 3$



$$S_1 = [2, +\infty) \cap (3, +\infty) = (3, +\infty)$$

2) $x+2 \leq 0$ y $x < 3$



$x \leq -2$ y $x < 3$

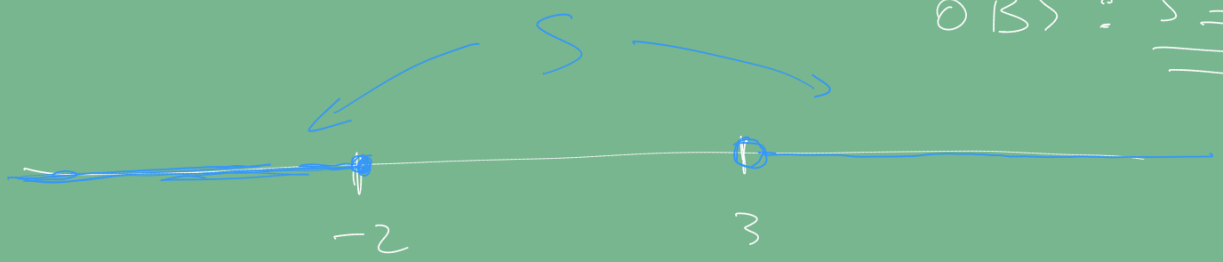
$$S_2 = (-\infty, -2] \cap (-\infty, 3) = (-\infty, -2]$$

LA SOLUCIÓN ES

$$S = S_1 \cup S_2 = (3, +\infty) \cup (-\infty, -2]$$

$$\boxed{S = (-\infty, -2] \cup (3, +\infty)}$$

OBS: $S = A$



EJ:

$$B = \left\{ x \in \mathbb{R} : \frac{2x-1}{x-3} < 1 \right\}$$

OBSERVAMOS QUE B SE REPRESENTA:

$-2 = \inf(A)$ B $3 = \sup(B)$



$\in$ -> "PERTENECE"

$$B = \{ x \in \mathbb{R} : x \notin A \} = A^c$$ ← COMPLEMENTO.

$$\frac{2x-1}{x-3} < 1$$

$$\frac{x+2}{x-3} < 0$$

$$\begin{cases} \frac{A}{B} \leq 0 \end{cases} \rightarrow \begin{cases} A \geq 0 \text{ y } B < 0 \\ A \leq 0 \text{ y } B > 0 \end{cases}$$

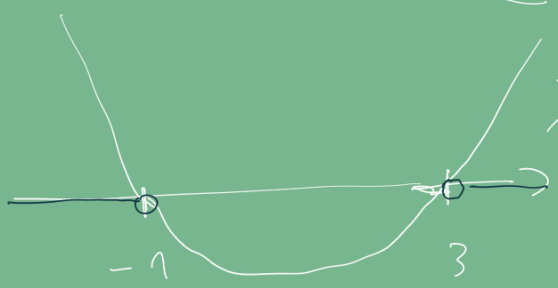
CASOS: 1) $x+2 > 0$ y $x-3 < 0$

2) $x+2 < 0$ y $x-3 > 0$

EJERCICIO: RESOLVER

OBS.: $x^2 - 2x - 3 > 0$

→ RAÍCES → $x = -1$, $x = 3$



$S = (-\infty, -1) \cup (3, +\infty)$

$1 \cdot x^2 - 2x - 3 = 1 \cdot (x+1) \cdot (x-3)$

$(x+1) \cdot (x-3) > 0$

CASO 1: $x + 1 > 0$ y $x - 3 > 0$

CASO 2: $x + 1 < 0$ y $x - 3 < 0$

$S = S_1 \cup S_2$ ← SOLUCIÓN

OBS: $A \cdot B \geq 0$ → $S = S_1 \cup S_2$

→ $A \geq 0$ y $B \geq 0$ ①

(o)

$A \leq 0$ y $B \leq 0$ ②

OBS.: $A \cdot B \leq 0$ → $S = S_1 \cup S_2$

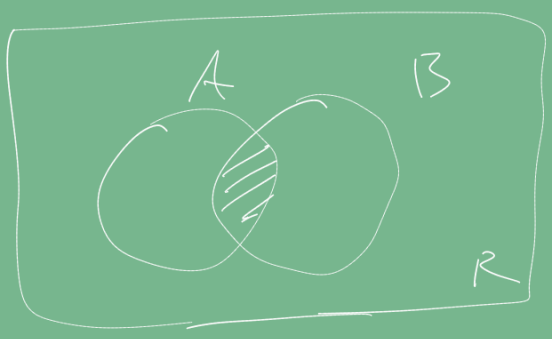
→ $A \geq 0$ y $B \leq 0$ ①

(o)

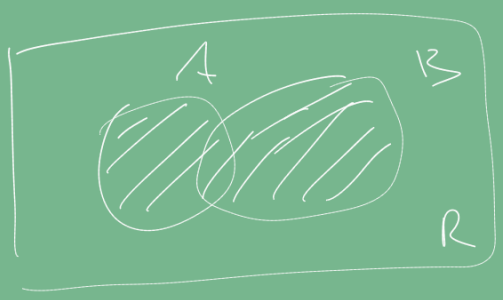
$A \leq 0$ y $B \leq 0$ ②

"y" → $\wedge$ $\quad\quad$ $\cap$ → "INTERSECCIÓN"

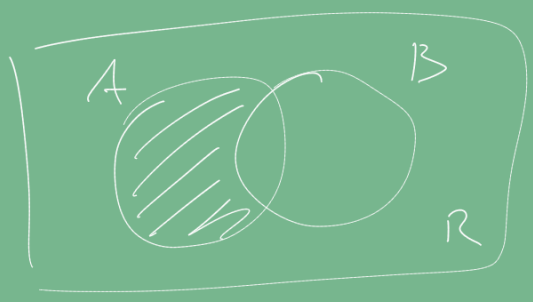
"o" → $\vee$ $\quad\quad$ $\cup$ → "UNIÓN"



$$A \cap B = \{x : x \in A \wedge x \in B\}$$



$$A \cup B = \{x \overset{x \in R}{:} x \in A \vee x \in B\}$$



$$A \setminus B = \{x \in R : x \in A \wedge x \notin B\}$$

$$A^c = R \setminus A = \{x \in R : x \notin A\}$$



OBS: $A \setminus B = A \cap B^c$